

Brian Jongwon Choi

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Nationality: USA

Interests	Partial differential equations, Harmonic analysis, Numerical analysis, Dynamical systems, Uncertainty quantification
Education	Boston University • PhD, Mathematics, May 2020 • Thesis (under Mark A Kon): <i>Weighted Fourier Analysis and Dispersive Equations</i> Boston College • B.S., Magna Cum Laude, Mathematics (Major), Physics (Minor), May 2014 • Recipient of the Sally Award 2014 (awarded to the top graduating math major) • Member of Pi Mu Epsilon (mathematics) and Sigma Pi Sigma (physics) • Thesis (under Dubi Kelmer): <i>An Algebraic Approach to Quantum Systems Using Finite Group Representation</i>
Appointment	University of Tennessee at Chattanooga • (Tenure-track) Assistant professor (Department of Mathematics), Jul2025-Present United States Military Academy West Point • Assistant Professor (Department of Mathematical Sciences), Jun 2023-Present • Deputy Director for Center for Data Science, Jun 2023-Present Southern Methodist University • RTG Postdoctoral Fellow (Mentor: Alejandro Aceves), Jan 2021-Jun 2023 Boston University • Research Assistant, Jun 2020-Dec 2020 Boston College • Instructor, MATH1004 Finite Probability and Applications, Sept 2020-Dec 2020
Awards/Grants	Civilian Service Commendation Medal , Department of the Army Sep 2025 Faculty Research Funds (competitive internal grant), USMA, West Point 2024 - 25 Science Policy Fellowship , Society for Industrial and Applied Mathematics 2025 - 26 Korean Honor Scholarship , Consulate General of Korea in USA Sept 2019 DAAD RISE Scholarship , Paul-Erich-Institut, Germany May-Aug 2014 Sally Award , Department of Mathematics, Boston College May 2014 Gilman International Scholarship , Institute of International Education May 2012
Grant Proposals Under Review	• National Science Foundation (NSF), DMS: Applied Mathematics. Title: <i>Continuum limits and rigorous numerics for dispersive waves with memory</i> Role: Principal Investigator (PI). • National Science Foundation (NSF), DMS: Computational Mathematics. Title: <i>Complex analytic regularity and uncertainty quantification for stochastic singular non-linear phase transitions</i> Role: co-Principal Investigator (co-PI). • Korean-American Scientists and Engineers Association (KSEA): Young Investigator Grants Title: <i>Rigorous numerics for complex systems: memory and uncertainty in non-linear PDEs</i> Role: Principal Investigator (PI).

Publications/ Preprints

(with Castrillon, Kon, Norton, Xu) Analytic regularity of strong solutions for the complexified stochastic nonlinear Poisson-Boltzmann equation *Computers & Mathematics with Applications* (2026), **204**, p.97-122, doi.org/10.1016/j.camwa.2025.12.003.

Nonexistence of traveling wave solutions in the fractional Rosenau-Hyman equation via homotopy perturbation method (2025) arXiv:2502.07810 (submitted to *International Journal of Differential Equations*)

Nonlinear excitation of ground states on nonlocal lattices, *Discrete Contin. Dyn. Syst. Ser. B* (2025), doi.org/10.3934/dcdsb.2025162

(With Parcell, Starling) Optimizing performance of real-time detection and classification for military personnel and weapons *The ITEA Journal* (2024), **45**(3), DOI: 10.61278/itea.45.3.1005

Dynamics of periodic fractional discrete NLS in the continuum limit, *Applicable Analysis* (2025), **104**(14), doi.org/10.1080/00036811.2025.2557997.

(With Norton, Xu, Kon, Castrillon) Complex analyticity of the nonlinear Poisson-Boltzmann equation for the interface problem with random domains, *Numerische Mathematik* (2025), doi.org/10.1007/s00211-025-01509-z.

(With Marstaller, Aceves) Localization and global dynamics in the long-range discrete nonlinear Schrödinger equation (2023) arXiv:2309.11395 (submitted to *Journal of Difference Equations and Application*)

(with Walton) Speed of propagation of fractional dispersive waves, *Communications on Pure and Applied Analysis* (2025), doi:10.3934/cpaa.2025094.

(with Aceves) Continuum limit of 2D fractional nonlinear Schrödinger equation *Journal of Evolution Equations* (2023), **23**(30), p.1-35, doi.org/10.1007/s00028-023-00881-3.

(with Aceves) Well-posedness of mixed nonlinear Schrödinger equation, *Partial Differential Equations in Applied Mathematics* (2022), **6**(c), p.1-11, 100406, doi.org/10.1016/j.padiff.2022.100406.

Multilinear weighted estimates and quantum Zakharov system, *Mathematical Modelling and Analysis* (2022), **27**(2), 342-359, doi.org/10.3846/mma.2022.15555.

Remark on the adiabatic limit of quantum Zakharov system, *Bulletin of the Malaysian Mathematical Sciences Society* (2022), **45**, 17111735, doi.org/10.1007/s40840-022-01272-6.

Small time behavior and summability for Schrödinger equation, *Graduate Journal of Mathematics* (2021), **6**(2), 9-21.

Current Projects

- From twisted DNLS to magnetic NLS: continuum limits and nodal standing waves
- Uncertainty quantification for nonlinear phase transitions

University Teaching

Academic Semester Teaching, *University of Tennessee at Chattanooga*

- MATH 3810: Quantum Information Science and Technology

AY 26-27

Academic Semester Teaching, *University of Tennessee at Chattanooga*

	• MATH 1730: Precalculus; MATH 3000: Proof and logic	AY 25-26
	Academic Semester Teaching , <i>United States Military Academy</i>	
	• MA364 ENG. Math	Spring 2025
	• MA487 Real analysis II, MA365 ENG. Math, MA389 Indep. study	Fall 2024
	• MA104 Calculus I-II (Combined)	Spring 2024
	• MA205 Multivariable calculus	Fall 2023
	Academic Semester Teaching , <i>Southern Methodist University</i>	Jan21 - May23
	• MATH1337,1338,3302 Calculus I-III, respectively	
	• STAT2331 Introduction to Statistical Methods	
	Academic Semester Teaching , <i>Boston College</i>	
	• MATH1004 Finite Probability and Applications	Fall 2020
	Academic Semester Teaching , <i>Boston University</i>	
	• MA113: Introduction to Statistics	Spring 2020
	Summer Instructor , <i>Boston University</i>	
	• MA115,116 Statistics I-II	2015
	• MA123 Calculus I	2016
	• MA226 Differential Equations	2017
	Teaching Fellow (graduate student) , <i>Boston University</i>	2014-2020
	• EK102 Linear Algebra for Engineers	
	• MA121,122 Calculus I for Social Sciences, Calculus II for Social Sciences	
	• MA123 Calculus I	
	• MA226 Differential Equations	
	• MA411 Advanced Calculus	
	• MA511 Analysis I	
	• MA569 Operations Research	
Mentorship	Undergraduate student advising , (Mentee: <i>Theodore Grimes, Patrick Cain</i>)	AY2025
	• USMA: MA389 (Indep. Study): Numerical methods in dynamical systems	
	Undergraduate student advising , (Mentee: <i>Karly Parcell</i>)	AY2024
	• USMA: MA489-499 (Senior thesis): image classification via CNN/YOLO; paper published in the ITEA journal	
	PhD student advising , (Mentee: <i>Austin Marstaller</i>)	AY2023
	• Co-advised his PhD dissertation at Southern Methodist University	
	PhD student advising , (Mentee: <i>Steven Walton</i>)	AY2022
	• Southern Methodist University: paper submitted: <i>Communications on Pure and Applied Analysis</i>	
	Undergraduate student advising , (Mentee: <i>William Graham</i>)	AY2020
	• Directed Reading Program at Boston University: <i>Stochastic Differential Equations (Evans)</i>	
Invited talks	Seminar talk, UKC2026, Orlando, FL	Aug 2026
	SIAM Annual meeting, Cleveland, OH	Jul 2026
	“Complex analyticity and sparse-grid UQ for stochastic nonlinear Poisson-Boltzmann equation”	
	Department colloquium, Quantum Center, UTC	Apr 2026
	“Magnetic flux and nonlinear quantum waves on a ring”	
	Department colloquium, Kennesaw State University	Apr 2026

“From lattices to waves: Bridging discrete and continuous dynamics in nonlinear dispersive systems”	
AMS spring southeastern sectional, Savannah, GA “Continuum convergence of twisted multicore fiber to magnetic NLS”	Mar 2026
Department seminar, Kennesaw State University “From Flach’s conjecture to long-range DNLS: excitation thresholds, ground states, and stability”	Mar 2026
Department colloquium, University of Tennessee at Chattanooga “Excitation thresholds and breather dynamics in multi-dimensional nonlocal lattices”	Nov 2025
AMS Fall Eastern Virtual Sectional Meeting “Speed of propagation in nonlocal dispersive equations”	Oct 2025
3rd UNCG Virtual PDE Conference “Unique continuation of fractional dispersive equations”	Oct 2025
UKC Technical Group Symposium, Atlanta, GA “Uncertainty quantification and regularity in the stochastic nonlinear Poisson-Boltzmann equation”	Aug 2025
SIAM Dynamical Systems, Denver, CO “Nonlinear excitations in nonlocal lattices”	May 2025
Seminar talk, University of Tennessee Chattanooga, TN “From nonlinear waves to quantum systems: analysis, uncertainty, and applications”	Apr 2025
Seminar talk, Creighton University, NE “Nonlinear Waves and Quantum Systems: Analysis, Uncertainty, and Beyond”	Feb 2025
Seminar talk, Villanova University, PA “Bridging Nonlinear Waves and Quantum Systems: Analytical Tools, Uncertainty Quantification, and Real-World Models”	Jan 2025
AMS Eastern Sectional Meeting, Albany, NY “Kuramoto oscillators with time delay and memory effects”	Oct 2024
SIAM Nonlinear Waves and Coherent Structure, Baltimore, MD “Long-range Interaction on Lattice”	Jun 2024
Invited Talk, PDE School (UC Berkeley), Berkeley, CA “Nonlocal Discrete Solitons”	Jun 2024
Poster, ICERM, Providence, RI “Time Evolution of the mixed-Fractional NLS”	Apr 2024
Invited Talk, AMS East Sectional (Howard University) “Modulational instability and continuum limit of periodic fractional NLS”	Apr 2024
Seminar Talk, CUNY Graduate Center “Nonlocal dispersive lattice dynamics and continuum limit”	Mar 2024

Invited Talk, AMS Special Session Mathematical Physics JMM “On Localization of Fractional Discrete Schrödinger Equation”	Jan 2024
Seminar Talk, SUNY New Paltz “Application of Fractional Calculus to Physics Via Differential Equation”	Oct 2023
SIAM NNP 2023, New Jersey Institute of Technology (Contributed Talk) “Localization of Discrete Fractional Schrödinger Dynamics”	Oct 2023
Great Lakes Mathematical Physics, Oberlin College (Contributed Talk) “Discrete to Continuum Dynamics of Fractional Nonlinear Schrödinger Equation”	Jun 2023
SIAM DS 2023, Oregon (Contributed Talk) “Continuum Limit of 2D Fractional Nonlinear Schrödinger Equation”	May 2023
12th Ohio River Analysis Meeting (Contributed Talk) “On the Compactness of Solution Support of Fractional Dispersive Equations”	Mar 2023
United States Military Academy West Point, New York Teaching Demonstration	Feb 2023
Research Talk, Navy Research Lab (Invited by Dr.Dey) “Bifurcation of Discrete Solitons of the Nonlinear Schrödinger Equation”	Jan 2023
SIAM 2022 TX-LA, University of Houston “Continuum Limit of 2D Fractional Nonlinear Schrödinger Equation”	Nov 2022
Computational Science Seminar, University of Texas at Dallas “Continuum Limit of 2D Fractional Nonlinear Schrödinger Equation”	Oct 2022
(Poster) SIAM Analysis of PDE 2022, Berlin, Germany (Virtual) “Dynamics of the Mixed-Fractional Nonlinear Schrödinger Equation”	Mar 2022
Hamilton Methods and Asymptotic Dynamics, ICERM “Quantum Zakharov System with the Periodic Boundary Condition”	Dec 2021
SIAM 2021 TX-LA, UTRGV (Minisymposium) “On Mixed Nonlinear Schrödinger Equation”	Nov 2021
Ohio River Analysis Meeting, University of Kentucky “Periodic Quantum Zakharov System”	Mar 2020
AMS Sectional Meeting, Tufts University “Fourth-Order Perturbation of Cubic Nonlinear Schrödinger Equation”	Mar 2020
Joint Mathematics Meetings (Contributed Paper Session) “Global Well-posedness and Modified Strichartz Estimates for the Fourth-Order Schrödinger Equation”	Jan 2020
SIAM Analysis of PDE (at La Quinta) “Global Well-Posedness of the Adiabatic Limit of Quantum Zakharov System in 1D”	Dec 2019
BU/Brown/UMASS Dynamics & PDE Seminar	Nov 2019

	“Pointwise convergence of Full Schrödinger Operator”	
	MSRI: Recent topics on well-posedness and stability of incompressible fluid and related topics	Jul 2019
	“A.E. Pointwise Convergence of Schrödinger Operator to the Identity”	
	Geometric and Harmonic Analysis, University of Connecticut	Mar 2019
	“Pointwise Convergence of Schroedinger Operator to the Identity”	
	Boston Graduate Math Colloquium, Boston College	Feb 2018
	“Nonlinear Smoothing in Nonlinear Schrödinger Equation”	
	BU Student Dynamics Seminar, Boston University	
	• “Wellposedness theory for the 1-dimensional Adiabatic Limit of Quantum Zakharov System” • “Expository Talk: Fourier Restriction” • “Difficulties in Extending Onsager’s Conjecture to a Bounded Domain” • “Time-Frequency Analysis and Carleson’s Theorem” • “Representation Theory Methods in Quantum Systems”	Feb 2019 Nov 2017 Feb 2017 Dec 2016 Nov 2016
Professional Service	Final judge , Interdisciplinary Contest in Modeling (ICM) Article review , Computers & Mathematics with Applications 203 PhD defense committee (SMU) , Austin Marstaller’s PhD defense Article review , Computational and Mathematical Biophysics 12(1) Bradley Omar Fellowship review , United States Military Academy Grant proposal review , Army Research Office Research Experience for Undergraduates (REU at SMU) , Article review , Communications in Contemporary Mathematics 23(4) Student Dynamical Systems Seminar at Boston University , Organizer	Apr 2025-2026 Feb 2025 Dec 2024 Apr 2024 Feb 2024 Jan 2024 Jul 2023 2020 2017-2019
Skills	Machine Learning Specialization (Coursera) (Stanford University, DeepLearning.AI, Credential Id: Q3JH7TCP34VD), Introductory C Programming Specialization (Coursera) (Duke University, Credential Id: GA6S3L9UMQYH)	Aug 2022 Sept 2022
Miscellaneous	Assistant officer-in-charge, Korean American Relations Club, USMA Violin II, Boston University All-campus orchestra Violin II, Boston College Orchestra	2024 - 2025 2017 - 2018 2010 - 2011